

A proof of Conjecture 00000005980

Same singular spectrum, different polar decomposition

1 The conjecture

Conjecture 00000005980 reads:

Definition: The singular spectrum and the eigenvector statistics are two layers. *Conjecture:* There exist two non-Hermitian models with the same singular spectrum but different eigenvector statistics, and the separation is realized by an explicit pair with the same spectrum but different polar decompositions.

The claim is existential. We prove it with explicit witnesses, and Lean formalises every step (`conjecture_5980`). The construction and the eigenvector statistic adapt the accepted solution of the eigenvalue analogue 00000005960 (ColdSmile, GPL-3.0).

2 Reading

- A *model* is a probability law on real 2×2 matrices (a PMF). It is *non-Hermitian* if every matrix in its support is non-Hermitian, i.e. $A^* \neq A$. We also require each model to be non-degenerate (not a Dirac mass).
- The *singular spectrum* of A is Mathlib's `LinearMap.singularValues` of the Euclidean operator of A : the singular values “are the square roots of the (necessarily non-negative) eigenvalues of the self-adjoint operator” T^*T [Wikipedia, *Singular value*], in decreasing order with multiplicity. Two models have the *same singular spectrum* if the push-forward laws of `singularValues` agree. We also show that the laws of the characteristic polynomial (the eigenvalues) agree.
- An *eigenvector statistic* is a function of the eigenvectors. We use the intrinsic, sign-free statistic $S(A) = \|\Pi_{E_7(A)} e_0\|^2$, the squared length of the orthogonal projection of e_0 onto the eigenvalue-7 eigenspace. For a simple eigenvalue it equals the squared first coordinate of any unit eigenvector. The models have *different eigenvector statistics* if the laws of S differ.
- A *polar decomposition* of A is $A = UP$ with U orthogonal and P positive semidefinite [Wikipedia, *Polar decomposition*] (`IsPolarDecomposition`). The *explicit pair* is $X \in \text{supp } A$ and $Y \in \text{supp } B$. We require X and Y to have the same singular values *and* the same eigenvalue spectrum, so both readings of “same spectrum” are covered. We require that every polar decomposition of X differs from every polar decomposition of Y in *both* factors, and that $S(X) \neq S(Y)$.

3 Construction

Let $c(t) = \frac{1-t^2}{1+t^2}$, $s(t) = \frac{2t}{1+t^2}$ and $R(t) = \begin{pmatrix} c & -s \\ s & c \end{pmatrix}$, so $R(t)^T R(t) = R(t)R(t)^T = I$ for all real t . Set

$$S_0 = \begin{pmatrix} 7 & 12 \\ 0 & -2 \end{pmatrix}, \quad F(t) = R(t) S_0 R(t)^T, \quad U_0 = \frac{1}{5} \begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix}, \quad P_0 = \frac{1}{5} \begin{pmatrix} 21 & 28 \\ 28 & 54 \end{pmatrix}.$$

Model A is the fair law on $\{F(0), F(1)\}$ and model B is the fair law on $\{F(\frac{1}{2}), F(-\frac{1}{2})\}$. The explicit pair is $X = F(0) = S_0$ and $Y = F(\frac{1}{2}) = \frac{1}{25} \begin{pmatrix} -113 & 216 \\ -84 & 238 \end{pmatrix}$.

Lemma 1 (`family_explicit`, `family_not_isHermitian`). $F(t)_{01} - F(t)_{10} = 12(c^2 + s^2) = 12$, so no $F(t)$ is Hermitian.

Lemma 2 (`charpoly_family`, `spectrum_family`). $\chi_{F(t)} = (X-7)(X+2)$ and $\text{spec } F(t) = \{7, -2\}$ for all t .

Proof. $\chi_{RNR^T} = \chi_{NR^T R} = \chi_N$ (`charpoly_mul_comm`), and $\chi_{S_0} = (X-7)(X+2)$. □

Lemma 3 (`singularValues_family`, `singularValues_seed`). Every $F(t)$ has singular values 14, 1 (and 0 beyond index 1).

Proof. $F(t)^T F(t) = R(t)(S_0^T S_0)R(t)^T$, so it has the characteristic polynomial of $S_0^T S_0$. Mathlib's `eigenvalues_eq_eigenvalues_iff` says that the sorted eigenvalues of a symmetric operator are determined by its characteristic polynomial. So all $F(t)$ have the singular values of S_0 . Here $S_0^T S_0 = \begin{pmatrix} 49 & 84 \\ 84 & 148 \end{pmatrix}$ has trace 197 and determinant 196, so its characteristic polynomial is $(X-196)(X-1)$. The sorted roots are $[196, 1]$ (`sort_roots_charpoly_eq_eigenvalues`), and their square roots are 14, 1. □

Lemma 4 (`eigenspaceSeven_family`, `statistic_family`). $E_7(F(t)) = \mathbb{R}(c(t), s(t))$, and hence $S(F(t)) = c(t)^2$.

Proof. $F(t)(c, s)^T = RS_0 e_0 = 7(c, s)^T$. Conversely, if $F(t)w = 7w$, then $-s(Fw - 7w)_0 + c(Fw - 7w)_1 = -9u - 2u(c^2 + s^2 - 1)$ with $u = -sw_0 + cw_1$, so $u = 0$ and $w = (cw_0 + sw_1)(c, s)$. The projection onto a unit line is $\langle v, e_0 \rangle v$, so $S = c^2$. □

Hence the law of S is $\frac{1}{2}\delta_1 + \frac{1}{2}\delta_0$ under A (since $c(0) = 1$ and $c(1) = 0$) and $\delta_{9/25}$ under B (since $c(\pm\frac{1}{2}) = \frac{3}{5}$). The two laws differ, while both singular-value laws are $\delta_{(14,1,0,\dots)}$ and both characteristic-polynomial laws are $\delta_{(X-7)(X+2)}$.

Lemma 5 (`isPolar_family`, `polar_unique`). $F(t) = U(t)P(t)$ with $U(t) = RU_0R^T$ orthogonal and $P(t) = RP_0R^T$ positive semidefinite. For an invertible matrix the polar decomposition is unique.

Proof. $U_0 P_0 = S_0$ and $U_0^T U_0 = I$. Also $21(21x^2 + 56xy + 54y^2) = (21x + 28y)^2 + 350y^2 \geq 0$, so $P_0 \succeq 0$, and conjugation preserves both properties. For uniqueness: if $A = UP$, then $P \cdot P = A^T A$. Positive square roots are unique (`CFC.mul_self_eq_mul_self_iff`), so P is determined. Because $\det A \neq 0$, P is invertible, which determines U . Here $\det F(t) = -14$. □

So X has the unique polar decomposition (U_0, P_0) , and Y has the unique polar decomposition $(\frac{1}{125} \begin{pmatrix} -117 & 44 \\ 44 & 117 \end{pmatrix}, \frac{1}{125} \begin{pmatrix} 381 & -592 \\ -592 & 1494 \end{pmatrix})$. Both factors differ, as the $(0,0)$ entries show. Finally, $S(X) = 1 \neq \frac{9}{25} = S(Y)$.

Theorem 6 (`explicit_separation`, `conjecture_5980`). *Separation $A B X Y$ holds. It asserts the following. Both models are supported on non-Hermitian matrices and are not Dirac. Their `singularValues` laws are equal, and their `charpoly` laws are equal. Their laws of S differ. $X \in \text{supp } A$ and $Y \in \text{supp } B$, with equal singular values and equal spectra. Polar decompositions of X and of Y exist, any two of them (one for X , one for Y) differ in both factors, and $S(X) \neq S(Y)$. Hence $\exists A B X Y$, Separation $A B X Y$.*

4 Lean formalisation and axiom audit

Lean 4 (`leanprover/lean4:v4.33.1`) with Mathlib v4.33.1; file `lean/Conjecture5980/Basic.lean` (412 lines, only import Mathlib). Main statement:

```
theorem conjecture_5980 :
  exists (A B : PMF Mat2) (X Y : Mat2), Separation A B X Y
```

Here `Mat2 = Matrix (Fin 2) (Fin 2) Real`. `singularValues A` is `(toEuclideanLin A).singularValues`, and `statistic A` is `norm ((eigenspaceSeven A).starProjection firstAxis) ^ 2`. `lake build` succeeds. `#print axioms` for `conjecture_5980`, `explicit_separation`, `singularValues_seed` and `polar_unique` each reports `[propext, Classical.choice, Quot.sound]`. There is no `sorry` and no `native_decide`.

Scope. The claim is existential, with no dimension-growth, independence or asymptotic hypothesis. We claim nothing about universality. The statistic is one specific eigenvector statistic. Eigenvector-overlap statistics are invariant under the orthogonal conjugation used here, so we do not claim that they separate the two models.

References

- [1] Wikipedia, *Singular value*, https://en.wikipedia.org/wiki/Singular_value.
- [2] Wikipedia, *Polar decomposition*, https://en.wikipedia.org/wiki/Polar_decomposition.